ASSIGNMENT-2

Objective This Assignment is related to the image classification

This Assignment contain '2' parts

① Provide the MNIST Data to the model and Detect the label.

② In MNIST there are total 10 classes i.e from 0 to 9

Now we are doing more deep classification that is

Hard written Class label 0 and 6 belong to Class 0.

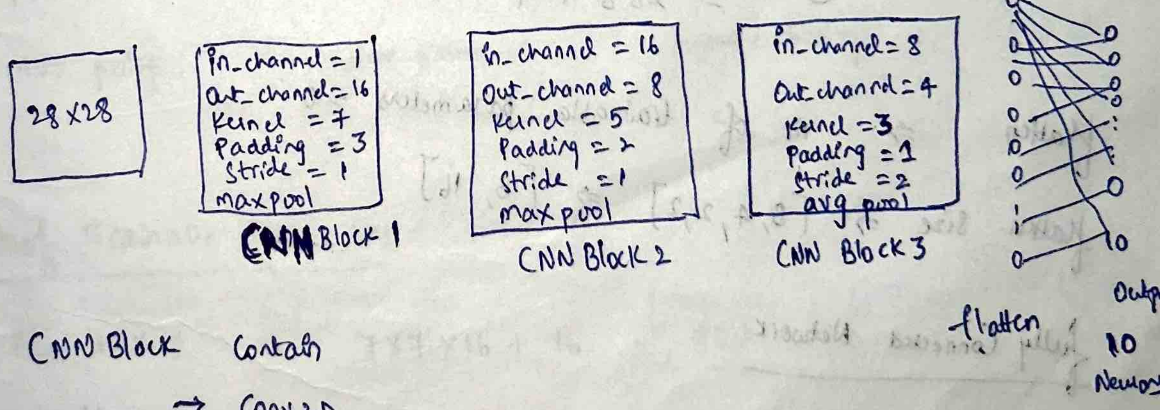
Hand written Class label 1 and 7 belong to Class 1

Hard written Class label 2, 3, 8, 5 belong to Class 2

Hard written Class label 4, 9 belong to Class 3

Architecture for part 1 of the Assignment

- The image size of the MNIST is 28 x28 . and It is Grey scale No of channels = 1

- 

- CNN Block Contain
  - → Conv 2D
  - → Activation Relu
  - → max pool (or) avg pool

- $[B, 1, 28, 28] \xrightarrow[\text{Block1}]{\text{CNN}} [B, 16, 14, 14] \xrightarrow[\text{Block 2}]{\text{CNN}} [B, 8, 7, 7] \xrightarrow[\text{Block 3}]{\text{CNN}} [B, 4, 2, 2]$

  $[B, 4, 2, 2] \xrightarrow{\text{flatten}} [B, 16] \xrightarrow[\text{layer}]{\text{Dense}} [B, 10]$

Trainable Parameters

CNN Block 1 $\Rightarrow$ Input to this Block is [B, 1, 28, 28]

kernel_size = 7 stride = 1 padding = 3

Out_channels = 16

So $\underbrace{7 \times 7 \times 16}_{\text{kernel parameter}} + \underbrace{16}_{\text{bias}}$

$= 784 + 16 = 800$

CNN Block 2 $\Rightarrow$ Input to this Block is [B, 16, 14, 14]

Kernel_size = 5 stride = 1 padding = 2

Out_channels = 8

So $\underbrace{5 \times 5 \times 16 \times 8}_{\text{kernel parameter}} + \underbrace{8}_{\text{bias}}$

$= 3200 + 8 = 3208$

CNN Block 3 $\Rightarrow$ Input of this Block is [B, 8, 7, 7]

kernel_size = 3 stride = 2 padding = 1

Out_channels = 4

So $3 \times 3 \times 8 \times 4 + 4$

$= 288 + 4 = 292$

flatten $\Rightarrow$ No of trainable parameters = 0

flatten size $\Rightarrow$ [B, 4, 2, 2] $\Rightarrow$ [B, 16]

fully Connected Network

$16 \times 10 + 10$

$= 170$

So $800 + 3208 + 292 + 0 + 170$

$= 4470$

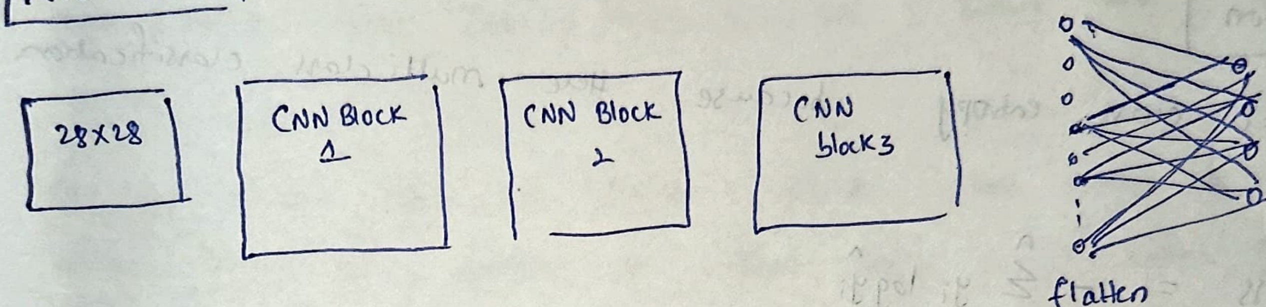
→ Train images = 50000

Test images = 10000

→ Test Accuracy = 98.57 %

PART 2 of Assignment

Architecture.



CNN block 1	CNN Block 2	CNN Block 3	fully Connected
in_channel = 1	in_channel = 16	in_channel = 8	Output layer
out_channel = 16	out_channel = 8	out_channel = 4	4 Neurons
kernel_size = 7	kernel_size = 5	Kernel_size = 3	because there
padding = 3	padding = 2	padding = 1	are only
Stride = 1	Stride = 1	Stride = 2	4
Relu activation	Relu activation	Relu activation	classes
max pooling	max pooling	Average pooling	

No of Trainable parameters

CNN block 1 => $7 \times 7 \times 16 + 16 = 800$

CNN block 2 => $5 \times 5 \times 16 \times 8 + 8 = 3208$

CNN block 3 => $3 \times 3 \times 8 \times 4 + 4 = 288 + 4 = 292$

flatten => 0

fully Connected => $16 \times 4 + 4 = 68$

$= 800 + 3208 + 292 + 0 + 68$

$= 4368$

Test Accuracy = 99.03 %

Parameters

batch size = 128

epochs = 20

learning-rate = 0.001

Loss function

Categorical cross entropy because Here multi class classification is done.

$$loss = -\sum_{i=1}^{n} y_i \log \hat{y}_i$$

Optimizer

Here we used the Adam Optimizer.

1. Compute Gradient

$$g_t = \nabla_\theta L(\theta_t)$$

g_t = gradient at time step t

L = Loss function

2. First moment estimate (momentum)

$$m_t = \beta_1 m_{t-1} + (1-\beta_1) g_t$$

m_t = first moment estimate

β_1 = momentum coefficient (typically 0.9)

3. Second moment estimate (RMS prop)

$$V_t = \beta_2 V_{t-1} + (1-\beta_2) g_t^2$$

V_t = Second moment estimate

β_2 = Variance coefficient (0.999)

(4) Bias Correction of First moment

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}$$

(5) Bias Correction of Second moment

$$\hat{v}_t = \frac{v_t}{1-\beta_2^t}$$

(6) Parameter Update

$$\theta_{t+1} = \theta_t - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t}} + \epsilon$$

$\alpha = 0.001$

$\epsilon = 10^{-8}$ (to avoid division by 0)